Lessons from Building Claude Code: How We Use Skills

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https://x.com/trq212/status/2033949937936085378

Skills have become one of the most used extension points in Claude Code. They're flexible, easy to make, and simple to distribute.

But this flexibility also makes it hard to know what works best. What type of skills are worth making? What's the secret to writing a good skill? When do you share them with others?

We've been using skills in Claude Code extensively at Anthropic with hundreds of them in active use. These are the lessons we've learned about using skills to accelerate our development.

What are Skills?

If you're new to skills, I'd recommend reading our docs or watching our newest course on new Skilljar on Agent Skills, this post will assume you already have some familiarity with skills.

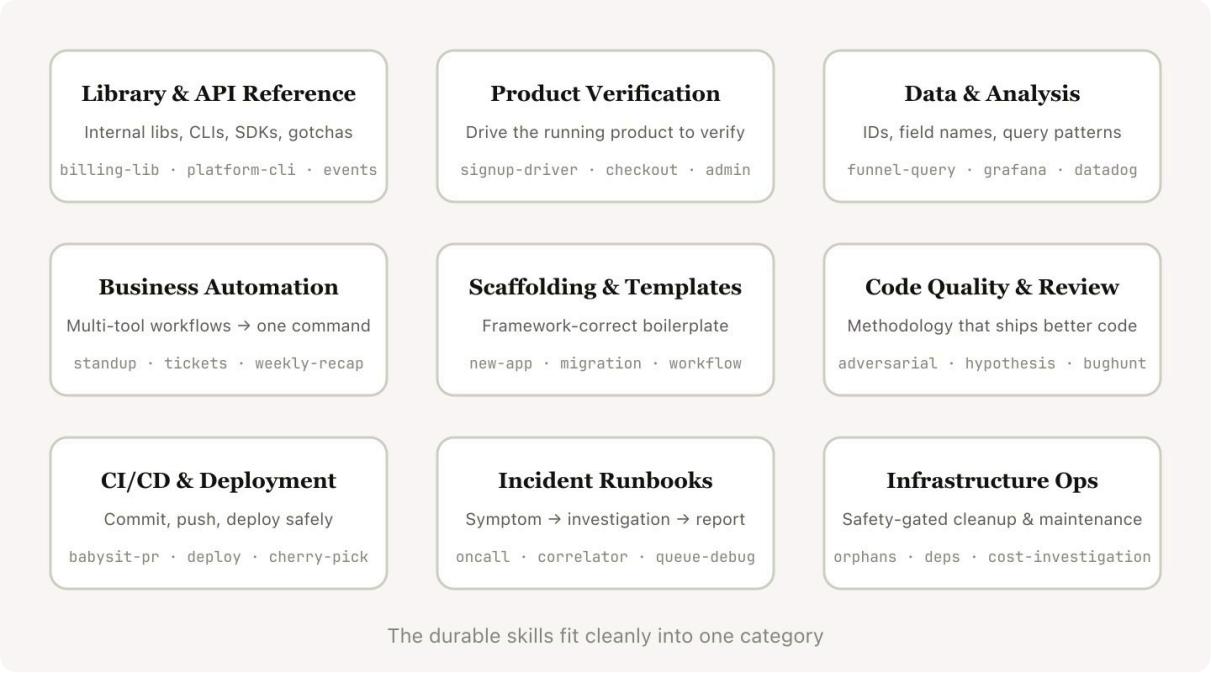
A common misconception we hear about skills is that they are "just markdown files", but the most interesting part of skills is that they're not just text files. They're folders that can include scripts, assets, data, etc. that the agent can discover, explore and manipulate.

In Claude Code, skills also have a wide variety of configuration options including registering dynamic hooks.

We've found that some of the most interesting skills in Claude Code use these configuration options and folder structure creatively.

Types of Skills

After cataloging all of our skills, we noticed they cluster into a few recurring categories. The best skills fit cleanly into one; the more confusing ones straddle several. This isn't a definitive list, but it is a good way to think about if you're missing any inside of your org.



1. Library & API Reference

Skills that explain how to correctly use a library, CLI, or SDKs. These could be both for internal libraries or common libraries that Claude Code sometimes has trouble with. These skills often included a folder of reference code snippets and a list of gotchas for Claude to avoid when writing a script.

Examples:

- billing-lib — your internal billing library: edge cases, footguns, etc.
- internal-platform-cli — every subcommand of your internal CLI wrapper with examples on when to use them
- frontend-design — make Claude better at your design system

2. Product Verification

Skills that describe how to test or verify that your code is working. These are often paired with an external tool like playwright, tmux, etc. for doing the verification.

Verification skills are extremely useful for ensuring Claude's output is correct. It can be worth having an engineer spend a week just making your verification skills excellent.

Consider techniques like having Claude record a video of its output so you can see exactly what it tested, or enforcing programmatic assertions on state at each step. These are often done by including a variety of scripts in the skill.

Examples:

- signup-flow-driver — runs through signup → email verify → onboarding in a headless browser, with hooks for asserting state at each step
- checkout-verifier — drives the checkout UI with Stripe test cards, verifies the invoice actually lands in the right state
- tmux-cli-driver — for interactive CLI testing where the thing you're verifying needs a TTY

3. Data Fetching & Analysis

Skills that connect to your data and monitoring stacks. These skills might include libraries to fetch your data with credentials, specific dashboard ids, etc. as well as instructions on common workflows or ways to get data.

Examples:

- funnel-query — "which events do I join to see signup → activation → paid" plus the table that actually has the canonical user_id
- cohort-compare — compare two cohorts' retention or conversion, flag statistically significant deltas, link to the segment definitions
- grafana — datasource UIDs, cluster names, problem → dashboard lookup table

4. Business Process & Team Automation

Skills that automate repetitive workflows into one command. These skills are usually fairly simple instructions but might have more complicated dependencies on other skills or MCPs. For these skills, saving previous results in log files can help the model stay consistent and reflect on previous executions of the workflow.

Examples:

- standup-post — aggregates your ticket tracker, GitHub activity, and prior Slack → formatted standup, delta-only
- create--ticket — enforces schema (valid enum values, required fields) plus post-creation workflow (ping reviewer, link in Slack)
- weekly-recap — merged PRs + closed tickets + deploys → formatted recap post

5. Code Scaffolding & Templates

Skills that generate framework boilerplate for a specific function in codebase. You might combine these skills with scripts that can be composed. They are especially useful when your scaffolding has natural language requirements that can't be purely covered by code.

Examples:

- new--workflow — scaffolds a new service/workflow/handler with your annotations
- new-migration — your migration file template plus common gotchas
- create-app — new internal app with your auth, logging, and deploy config pre-wired

6. Code Quality & Review

Skills that enforce code quality inside of your org and help review code. These can include deterministic scripts or tools for maximum robustness. You may want to run these skills automatically as part of hooks or inside of a GitHub Action.

- adversarial-review — spawns a fresh-eyes subagent to critique, implements fixes, iterates until findings degrade to nitpicks
- code-style — enforces code style, especially styles that Claude does not do well by default.
- testing-practices — instructions on how to write tests and what to test.

7. CI/CD & Deployment

Skills that help you fetch, push, and deploy code inside of your codebase. These skills may reference other skills to collect data.

Examples:

- babysit-pr — monitors a PR → retries flaky CI → resolves merge conflicts → enables auto-merge
- deploy- — build → smoke test → gradual traffic rollout with error-rate comparison → auto-rollback on regression
- cherry-pick-prod — isolated worktree → cherry-pick → conflict resolution → PR with template

8. Runbooks

Skills that take a symptom (such as a Slack thread, alert, or error signature), walk through a multi-tool investigation, and produce a structured report.

Examples:

- -debugging — maps symptoms → tools → query patterns for your highest-traffic services
- oncall-runner — fetches the alert → checks the usual suspects → formats a finding
- log-correlator — given a request ID, pulls matching logs from every system that might have touched it

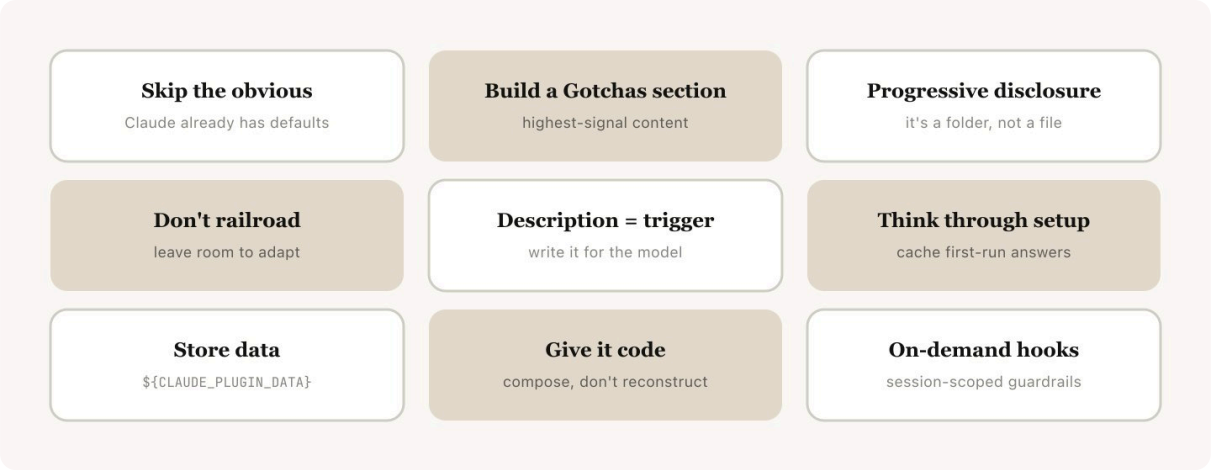
9. Infrastructure Operations

Skills that perform routine maintenance and operational procedures — some of which involve destructive actions that benefit from guardrails. These make it easier for engineers to follow best practices in critical operations.

Examples:

- -orphans — finds orphaned pods/volumes → posts to Slack → soak period → user confirms → cascading cleanup
- dependency-management — your org's dependency approval workflow
- cost-investigation — "why did our storage/egress bill spike" with the specific buckets and query patterns

Tips for Making Skills



Once you've decided on the skill to make, how do you write it? These are some of the best practices, tips, and tricks we've found.

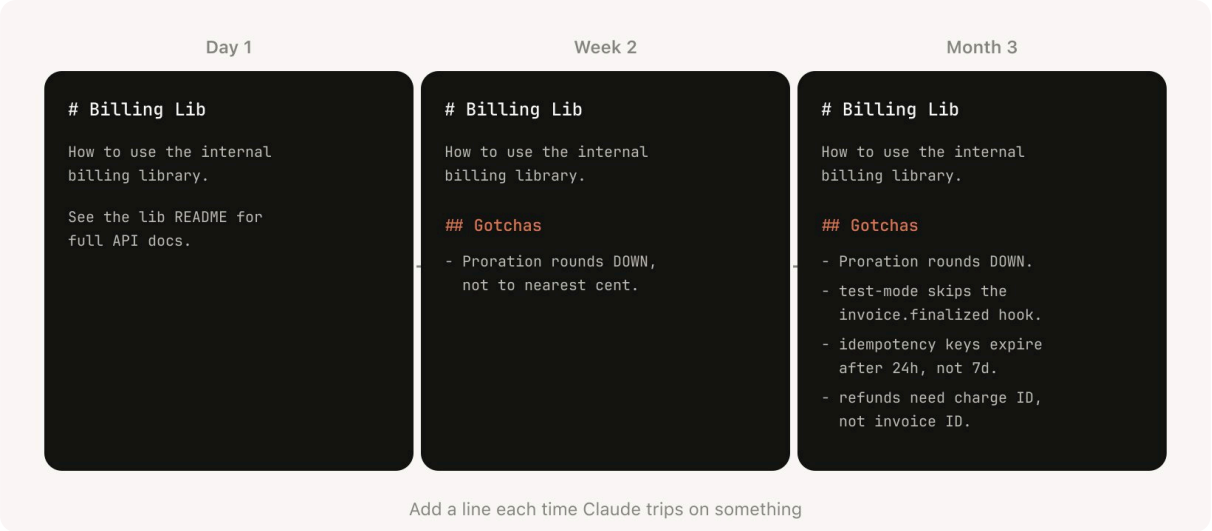
We also recently released Skill Creator to make it easier to create skills in Claude Code.

Don’t State the Obvious

Claude Code knows a lot about your codebase, and Claude knows a lot about coding, including many default opinions. If you’re publishing a skill that is primarily about knowledge, try to focus on information that pushes Claude out of its normal way of thinking.

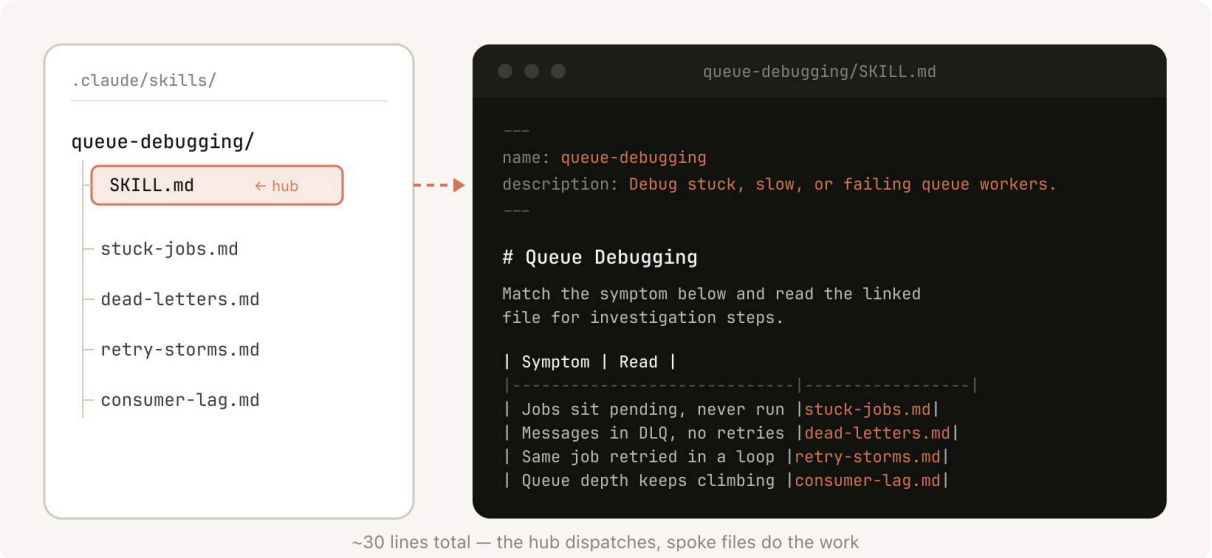
The frontend design skill is a great example — it was built by one of the engineers at Anthropic by iterating with customers on improving Claude’s design taste, avoiding classic patterns like the Inter font and purple gradients.

Build a Gotchas Section



The highest-signal content in any skill is the Gotchas section. These sections should be built up from common failure points that Claude runs into when using your skill. Ideally, you will update your skill over time to capture these gotchas.

Use the File System & Progressive Disclosure



Like we said earlier, a skill is a folder, not just a markdown file. You should think of the entire file system as a form of context engineering and progressive disclosure. Tell Claude what files are in your skill, and it will read them at appropriate times.

The simplest form of progressive disclosure is to point to other markdown files for Claude to use. For example, you may split detailed function signatures and usage examples into

references/api.md.

Another example: if your end output is a markdown file, you might include a template file for it in assets/ to copy and use.

You can have folders of references, scripts, examples, etc., which help Claude work more effectively.

Avoid Railroading Claude

Claude will generally try to stick to your instructions, and because Skills are so reusable you'll want to be careful of being too specific in your instructions. Give Claude the information it needs, but give it the flexibility to adapt to the situation. For example:

Too prescriptive

```
Step 1: Run git log to find the commit.
Step 2: Run git cherry-pick <hash>.
Step 3: If there are conflicts, run git status to list them.
Step 4: Open each conflicting file.
Step 5: For each <<< marker, decide which side to keep.
Step 6: Run git add on each resolved file, then…
```

Better

```
Cherry-pick the commit onto a clean branch. Resolve conflicts
preserving intent. If it can't land cleanly, explain why.
```

Think through the Setup

```
standup-post/SKILL.md

---
name: standup-post
description: Post your daily standup. Triggers on "standup", "daily".
---

## Your config

!`cat ${CLAUDE_SKILL_DIR}/config.json 2>/dev/null || echo "NOT_CONFIGURED"`

## Instructions

If the config above is NOT_CONFIGURED, ask the user:
- Which Slack channel?
- Paste a sample standup you liked
Then write the answers to ${CLAUDE_SKILL_DIR}/config.json.

Otherwise, post to the saved channel using the saved format.
```

The !`...` line runs as a shell command before Claude reads the prompt

Some skills may need to be set up with context from the user. For example, if you are making a skill that posts your standup to Slack, you may want Claude to ask which Slack channel to post it in.

A good pattern to do this is to store this setup information in a config.json file in the skill directory like the above example. If the config is not set up, the agent can then ask the user for information.

If you want the agent to present structured, multiple choice questions you can instruct Claude to use the AskUserQuestion tool.

The Description Field Is For the Model

When Claude Code starts a session, it builds a listing of every available skill with its description. This listing is what Claude scans to decide "is there a skill for this request?" Which means the description field is not a summary — it's a description of when to trigger this PR.

```
SKILL.md
---
name: babysit-pr
description: A comprehensive tool for
monitoring pull request status across
the development lifecycle.
---
```

```
SKILL.md
---
name: babysit-pr
description: Monitors a PR until it
  merges. Trigger on 'babysit',
  'watch CI', 'make sure this lands'.
---
```

Memory & Storing Data

```
standup-post/SKILL.md

## Memory

Append each standup to ${CLAUDE_PLUGIN_DATA}/standups.log
after posting. This folder persists across skill upgrades.

---

On each run:
- read the log to see what changed since yesterday
- write today's entry after sending to Slack
```

Some skills can include a form of memory by storing data within them. You could store data in anything as simple as an append only text log file or JSON files, or as complicated as a SQLite database.

For example, a standup-post skill might keep a standups.log with every post it's written, which means the next time you run it, Claude reads its own history and can tell what's changed since yesterday.

Data stored in the skill directory may be deleted when you upgrade the skill, so you should store this in a stable folder, as of today we provide `${**CLAUDE_PLUGIN_DATA**}` as a stable folder per plugin to store data in.

Store Scripts & Generate Code

One of the most powerful tools you can give Claude is code. Giving Claude scripts and libraries lets Claude spend its turns on composition, deciding what to do next rather than reconstructing boilerplate.

For example, in your data science skill you might have a library of functions to fetch data from your event source. In order for Claude to do complex analysis, you could give it a set of helper functions like so:

```
lib/signups.py

def fetch(day):
    """Signups from events.raw for one day.
       - event='signup_completed', NOT 'signup_started'
       - dedupe by anonymous_id — user_id is null until after signup"""

def by_referrer(df):
    """Group by traffic source.
       - '(direct)' and '' and None all mean organic"""

def by_landing_page(df):
    """Group by entry page.
       - '/', '/index', '/home' are all the homepage
       - strips query params so UTM'd links collapse"""
```

Claude can then generate scripts on the fly to compose this functionality to do more advanced analysis for prompts like “What happened on Tuesday?”

```
investigate.py · generated by Claude

from lib.signups import fetch, by_referrer, by_landing_page

mon, tue = fetch("2024-03-11"), fetch("2024-03-12")

print(by_referrer(tue) - by_referrer(mon))      # organic -60%, paid flat
print(by_landing_page(tue) - by_landing_page(mon))  # homepage specifically

# → something broke on / on Tuesday
```

On Demand Hooks

Skills can include hooks that are only activated when the skill is called, and last for the duration of the session. Use this for more opinionated hooks that you don’t want to run all the time, but are extremely useful sometimes.

For example:

- /**careful** — blocks rm -rf, DROP TABLE, force-push, kubectl delete via PreToolUse matcher on Bash. You only want this when you know you're touching prod — having it always on would drive you insane
- /**freeze** — blocks any Edit/Write that's not in a specific directory. Useful
- when debugging: "I want to add logs but I keep accidentally 'fixing' unrelated

Distributing Skills

One of the biggest benefits of Skills is that you can share them with the rest of your team.

There are two ways you might to share skills with others:

- check your skills into your repo (under ./.claude/skills)
- make a **plugin** and have a Claude Code Plugin marketplace where users can upload and install plugins (read more on the documentation here)

For smaller teams working across relatively few repos, checking your skills into repos works well. But every skill that is checked in also adds a little bit to the context of the model. As you scale, an internal plugin marketplace allows you to distribute skills and let your team decide which ones to install.

Managing a Marketplace

How do you decide which skills go in a marketplace? How do people submit them?

We don't have a centralized team that decides; instead we try and find the most useful skills organically. If you have a skill that you want people to try out, you can upload it to a sandbox folder in GitHub and point people to it in Slack or other forums.

Once a skill has gotten traction (which is up to the skill owner to decide), they can put in a PR to move it into the marketplace.

A note of warning, it can be quite easy to create bad or redundant skills, so making sure you have some method of curation before release is important.

Composing Skills

You may want to have skills that depend on each other. For example, you may have a file upload skill that uploads a file, and a CSV generation skill that makes a CSV and uploads it. This sort of dependency management is not natively built into marketplaces or skills yet, but you can just reference other skills by name, and the model will invoke them if they are installed.

Measuring Skills

To understand how a skill is doing, we use a PreToolUse hook that lets us log skill usage within the company (example code here). This means we can find skills that are popular or are undertriggering compared to our expectations.

Conclusion

Skills are incredibly powerful, flexible tools for agents, but it's still early and we're all figuring out how to use them best.

Think of this more as a grab bag of useful tips that we've seen work than a definitive guide. The best way to understand skills is to get started, experiment, and see what works for you. Most of ours began as a few lines and a single gotcha, and got better because people kept adding to them as Claude hit new edge cases.

I hope this was helpful, let me know if you have any questions.